Bihar Engineering University, Patna
End Semester Examination -2023
Semester-VI

Course: B.Tech. | *Time: 03 Hours*
Code: 100607 | *Subject: Introduction to VLSI Design* | *Full Marks: 70*

Instructions:-
(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q1 *Choose the correct answer of the following (Any seven question only):* [2x 7 = 14]

(i) What are the basic processes involved in fabricating ICs using planar technology?
(ii) What is the purpose of oxidation process in IC fabrication?
(iii) What are the advantages of pnp transistor over npn transistor in IC technology.
(iv) What is positive & negative photoresists?
(v) What is the function of SiO_2 layer in an IC ?
(vi) What is the advantage of using dry etching process?
(vii) Why aluminum is preferred for metallization?
(viii) What are the advantages of ion implantation technique?
(ix) List out the steps used in the preparation of Si – wafers.
(x) What is meant by reactive plasma etching?

Long type question

1. Explain various steps involved in IC fabrication.

2. (a) What is masking. [4]
   (b) Explain lithography process in detail. [10]

3. (a) Define Etching. what are the types of Etching also write difference between wet etching and Dry Etching. [7]
   (b) Explain Diffusion process in IC fabrication. [7]

4. (a) Compare Ion implantation with diffusion. [7]
   (b) Implement Y= AB + $\overline{AC}$ + $A\bar{B}C$ using CMOS TG. [7]

5. Explain the process of epitaxial growth IC fabrication with neat diagram?. [14]

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6. Explain the realization of monolithic Integrated circuits? [14]

7. Explain the basic process used in silicon planar technology with neat diagram? [14]

8. Design XOR Gate using COMS. [14]

9. Explain the fabrication of MOSFET in an IC? [14]

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